

Vocabulary

Max / min: the turning point of a parabola (the vertex); opens up gives a minimum, opens down gives a maximum.

Increasing / decreasing: read left to right — the graph goes up (increasing) or down (decreasing).

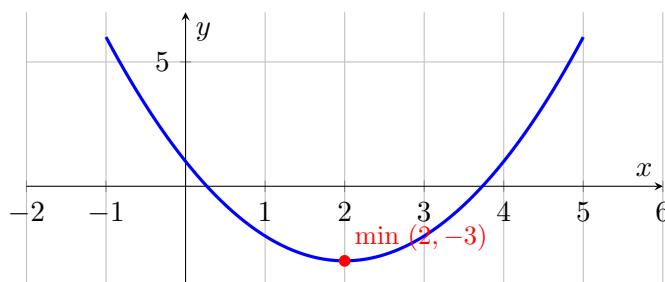
We do together. For $y = (x - 2)^2 - 3$, give the vertex, state max or min, and the intervals.

Step 1. Vertex form $a(x - h)^2 + k$ gives vertex (2, -3).

Step 2. Leading coefficient is +1 (opens up), so the vertex is a minimum.

Step 3. It decreases for $x < 2$ and increases for $x > 2$.

See it:



opens up $\Rightarrow$ minimum at $(2, -3)$.

You Try. Solve each. Show every step, then name the answer.

a) Give the vertex and max/min of $y = -x^2 + 4$.

vertex $(0, 4)$, maximum

b) Give the intervals of $y = x^2$.

decreasing $x < 0$, increasing $x > 0$

Summary (fill in): Opens up $\Rightarrow$ minimum; opens down $\Rightarrow$ maximum; the graph turns at the vertex.